

Lessons from a closed programme

A post-mortem of Saturated Superfluid Vacuum theory

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Abstract

The Saturated Superfluid Vacuum (SSV) programme — an attempt to derive the particle, force, gravity, and cosmology sectors from a single logarithmic Schrödinger order parameter with the vacuum treated as a real superfluid medium — is closed. No surviving route delivers the quantities it set out to derive. This is a post-mortem, written to the programme’s own standing rule that only correctly negative results are proof. Its purpose is to extract what generalises. The recurring diagnosis is *form yes, magnitude no*: the framework reproduces the kinematic *forms* of analogue and emergent gravity — weak-field time dilation, area-law horizon entropy, the Unruh/Hawking temperature, a logarithmic flat-rotation tail — essentially for free, because those follow from locality and symmetry, while every dynamical *magnitude* (Newton’s constant G , the cosmological scale, the particle masses) had to be conceded as an input. The most durable outputs are two negatives: a structural no-go — a scalar/acoustic vacuum cannot carry a spin-2 graviton or tensor gravitational-wave polarisations — and a concrete mode-counting demonstration that a literal holographic screen undershoots G by roughly 38 orders of magnitude. We also record the methodology that made clean failure possible, and the honest home of the surviving ontology (foundations / philosophy of physics). The lessons transfer beyond SSV to emergent-gravity and superfluid-vacuum model-building generally.

1 What this document is

A research programme can end in two ways: it can be abandoned, or it can be *closed* — worked until the reasons it cannot succeed are themselves results. SSV is closed in the second sense. The point of writing the ending down is that the reasons are more transferable than the framework was. Almost everything of value SSV produced is either a re-derivation of results already owned by the analogue- and emergent-gravity literature, or a negative; and the negatives are sharp enough to be useful to anyone building a vacuum out of a condensate. This note collects them, states the failures at full strength, and draws the lessons. It supersedes nothing; it concludes.

2 The programme, in brief

SSV proposed that the physical vacuum is a saturated superfluid described by a macroscopic order parameter $\Psi = \sqrt{\rho} e^{i\theta}$, that the fundamental particles are topological defects (vortices, breathers) in Ψ , and that the four forces, gravity, and cosmology all follow from one master dynamics — a logarithmic Schrödinger equation (LogSE) with a chiral-shear term and a saturation potential. The ambition was unusually total: from m_e and α alone, to derive the mass spectrum, the force sector, Newton’s constant, the arrow of time, black-hole structure,

galactic rotation curves, and the cosmological constant. The logarithmic nonlinearity is itself not new — it is the Bialynicki-Birula–Mycielski equation, developed as a superfluid vacuum by Zloschastiev — and the superfluid-vacuum lineage runs through Volovik. SSV’s distinctive bet was the totality: one equation, two inputs, everything else derived.

3 The method that let it fail cleanly

Before the failures, the part worth keeping. The programme ran under a discipline that is, in retrospect, its most exportable product:

- **A negative-results rule.** Only correctly negative results count as proof; positive results are merely suggestive and are never softened or buried. This single rule is what kept the programme from drifting into the permanent “suggestive” state that absorbs most heterodox physics.
- **Pre-registration.** Each hypothesis and its decision rule were fixed in a tracked issue *before* the computation ran, so a result could not be reinterpreted into a success after the fact.
- **Claim-status tracking.** Every quantitative claim carried an explicit status — *derived*, *coincidence*, or *falsified* — and a change of status had to propagate to the prose, the tables, and the record in one pass.
- **Computation-backed analysis.** Analytic claims were backed by a tested script plus a result note, so a claimed derivation had an executable witness and a reference could not silently break.

This machinery is why the programme could kill its own central claims rather than defend them. The lesson generalises: the value of pre-registration and a hard negative-results rule is highest precisely where external peer pressure is lowest.

4 What was reproduced: the form

In its gravity sector SSV reproduced, correctly, a sequence of standard results. Identifying local time with the order-parameter phase rate via the Josephson relation $\hbar\dot{\theta} = -\mu$ gives an emergent lapse, and weak-field this yields $g_{00} = 1 + 2\Phi/c^2$ with the Einstein-1911 (time-delay) half of light deflection. The acoustic horizon — the surface where the background flow turns sonic — behaves as a black-hole horizon with a definite surface gravity, no interior singularity (the finite vortex-core size cuts off collapse), and an Unruh/Hawking temperature of the right form. Quantising the LogSE Bogoliubov fluctuations gives an area-law horizon entanglement entropy. A logarithmic circulation-halo potential $\Phi_h = v_h^2 \ln r$ produces a flat rotation-curve tail, and ordinary Newtonian self-gravity of the baryons grows spiral arms and a bar by swing amplification, with no central black hole and no dark-matter particle.

Every item in that list is a known result of analogue/emergent gravity (Unruh, Visser, Volovik, Jacobson) or of standard galactic dynamics. SSV re-derived them inside one medium, which has pedagogical value, but none is new. This matters for the diagnosis: the *forms* came easily.

5 What failed: magnitudes and structural no-gos

5.1 The particle masses became inputs and coincidences

The mass sector reduced to two inputs, m_e and α , rather than deriving a spectrum. The “ α -harmonic mass ladder” was demoted to a numerical coincidence, not a derived spectrum. The muon as a derived breather mode failed three independent null tests and was closed. The W mass *scale* $m_W \sim m_e/\alpha^2$ is structurally derived, but the prefactor that brings it to the measured value is an $\mathcal{O}(1)$ coincidence. The honest tally: the sector reproduces scales it is handed and fits the rest.

5.2 The scalar-gravity no-go (the durable negative)

The sharpest structural result is a no-go. SSV’s gapless carrier is a single scalar phonon; the emergent acoustic metric is conformal-in-space plus a vector flow term, with *no* transverse-traceless part. So the framework’s gravitational radiation is helicity-0, the tensor overlap with a quadrupole source is zero, and the isotropic part cancels in a differential readout. The attempt to rescue a tensor mode from the spinor sector fails for a representation-theoretic reason: $\bar{\mathbf{2}} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3}$ contains no spin-2. The same square-root-of-a-vector fact that gives the photon spin-1 *forbids* a spin-2 graviton. SSV’s gravity is therefore *structurally scalar*: it can mimic light bending in the conformal/trace sector but cannot be general relativity, and its predicted gravitational-wave polarisation content is wrong (the original “effective spin-2” claim was falsified and withdrawn; the surviving model passes the Hulse–Taylor test only because a scalar emits no dipole). This is the Visser kinematics/dynamics wall, stated explicitly: an acoustic metric supplies geometry, not Einstein dynamics. It is a clean, transferable constraint on *any* scalar/superfluid emergent-gravity model.

5.3 Newton’s constant: a screen too sparse by 38 orders

Entropic and holographic gravity (Jacobson, Verlinde, Padmanabhan) keep the holographic screen abstract: one *assumes* an area law with the Planck-scale coefficient. SSV’s distinctive move was to make the screen literal — a real acoustic horizon in a real medium — so that the coefficient becomes computable rather than assumed. The form survived: the horizon entanglement entropy is area-law, computed by the correlator method (Peschel; Casini–Huerta) via the Srednicki radial decomposition, with the chiral term exactly silent. But this is the cheap half: the area law follows from the LogSE quadratic Hamiltonian being *local*, generic for any local free field (Srednicki; Bombelli–Koul–Lee–Sorkin; Jacobson’s entanglement-equilibrium). It *earns* an area law the papers had merely asserted; it does not surprise.

The coefficient is the test, and it fails. Closing the Jacobson construction with the computed $dS = \eta dA$ gives $\nabla^2 \Phi = 4\pi G_{\text{eff}} e/c^2$ with $G_{\text{eff}} = 1/(4\hbar c \eta)$. Reproducing the measured G requires

$$N = \frac{1}{\alpha_G} = \frac{\hbar c}{G m_p^2} = \left(\frac{m_P}{m_p} \right)^2 \simeq 1.7 \times 10^{38} \quad (1)$$

microstates per grain — the Planck-to-proton hierarchy. The substrate supplies $\mathcal{O}(10)$ localised Bogoliubov modes per healing-length cell, a few bits. The literal screen is short by about 10^{37} .

When a holographic screen is given a real microscopic substrate and its localised degrees of freedom are *counted*, the area law emerges for free but the Planck-scale coefficient does not:

the count falls short by ~ 38 orders. G must enter below the effective theory’s cutoff — a trans-Planckian input, not an emergent output.

This is the Sakharov induced-gravity problem and the Susskind–Uglum species problem, which Volovik names as the structural weak point of superfluid gravity — here *demonstrated by direct mode-counting* rather than argued. (Guard: the identity $(a_p/\ell_P)^2 = 1/\alpha_G$ merely restates the hierarchy and must not be read backwards as deriving N ; the negative is the *measured* count of $\mathcal{O}(10)$ confronted with the target.)

5.4 The arrow of time: a coarse-graining, not a resolution

SSV originally read its irreversibility as something new. A pre-registered reversal-echo computation through a topological-change event settled it the other way: the master dynamics is conservative and exactly time-reversal symmetric, and exact reversal recovers the pre-event state (fidelity deficit $1 - F \approx 8.7 \times 10^{-12}$ in 2D annihilation, $F = 0.999997$ for 3D Hopf-link reconnection, the deficit shrinking under refinement). The account of irreversibility is therefore a *coarse-graining* one in Boltzmann’s lineage, with the forgotten variables named concretely (phonon phases, Kelvin waves, vortex tangle, correlations); it is not a new resolution of Loschmidt’s paradox, and the programme withdrew any such claim. The genuinely open half — the low-entropy past — was never supplied. (The reconnection-reversibility computation does survive as a stand-alone result for the quantum-turbulence literature, shorn of the arrow-of-time framing.)

5.5 The cosmological acceleration: survives but underdetermined

The MOND-like scale was predicted as $a_0 = cH_0/2\pi$ from a de Sitter Gibbons–Hawking 2π , the same factor used for local horizons; the cosmological-constant alternative $c^2\sqrt{\Lambda}$ was excluded at $8.4\times$. But the data do not select the coefficient: SSV’s own halo estimator favours $1/2\pi$ while the independent radial-acceleration relation mildly favours Verlinde’s $1/6$, and the two straddle by $\sim 6\%$. The prediction is consistent and falsifiable (the decisive separator is high-redshift evolution) but unconfirmed, and the absolute scale is conceded in parallel with G .

6 The diagnosis: form yes, magnitude no

One pattern runs through every sector. SSV reproduces the *form* of the right answer cheaply, and concedes the *magnitude*. Time dilation: form yes, G no. Area law: form yes, coefficient no. Hawking temperature: form yes, factor of four no. Flat rotation curves: form yes, halo amplitude not derived. Cosmological a_0 : form yes, scale conceded. Masses: scales handed in, the rest fitted.

The reason is not bad luck; it is structural, and it is the real lesson. Kinematic forms — metric structure, area laws, dispersion relations, conservation laws — follow from *locality and symmetry*, which an emergent medium inherits almost automatically. The dynamical magnitudes — the coupling constants — live in the *microscopic* theory below the effective cutoff. In a superfluid vacuum the induced gravitational coupling is trans-Planckian by construction (Volovik), so the effective theory that produces the beautiful forms is structurally the wrong place to compute the constants. An emergent-gravity model gets geometry for free and the gravitational constant never. Recognising this early would have saved years; it is the single most useful thing the programme learned.

7 The surviving ontology and its honest home

What remains attractive is the ontology: local time as the rate of phase update of the medium; a finite update capacity that loading consumes, slowing the local clock; and gravity as the response of a holographic record to that loading. Each component already has a mainstream owner — time-as-phase-rate is the Josephson relation plus analogue g_{00} ; finite update capacity is the Margolus–Levitin/Lloyd bound on the rate of state change; time-as-change is the relational tradition (Barbour; Connes–Rovelli); screen-update-as-gravity is Jacobson/Verlinde/Padmanabhan. The ontology adds interpretive coherence (one phase field carrying the local clock, the static near-field, and the long-range encoding) but no prediction those programmes lack.

Two honest caveats belong with it. First, an internal tension: the bulk-realist reading (the medium’s local state slows time) and the holographic reading (“nothing bends space in the bulk; geometry is localised into the encoding”) are reconciled only by a duality claim that carries no falsifier of its own; it is a heuristic frame, not a mechanism. Second, the one component that does pull weight is *presentism*: because the emergent metric is effective and need not be geodesically complete, there is no real future infinity, which dissolves the obstacle that breaks global de Sitter holography and routes naturally to observer / static-patch holography (Chandrasekaran–Longo–Penington–Witten). But that removes an obstacle without building the dual: the present cosmological horizon still has a Gibbons–Hawking entropy whose microstates the medium does not supply — the same magnitude gap as everywhere else. The honest home for the ontology is therefore foundations / philosophy of physics, not a theory of gravity.

8 Lessons that generalise

1. **Forms are cheap, constants are dear.** In any emergent/analogue programme, expect to reproduce kinematic forms from locality and symmetry, and expect the coupling constants to require microscopic input the effective theory structurally cannot provide. Treat a reproduced *form* as confirmation of consistency, never as a derivation of the *magnitude*.
2. **Negatives are the durable output.** The scalar-gravity no-go and the mode-counting G -shortfall will outlast everything positive the programme proposed, precisely because a correct negative generalises and a numerical coincidence does not.
3. **Concreteness exposes what abstraction hides.** Abstract entropic gravity conceals the magnitude problem; building the screen out of real degrees of freedom and counting them makes the 10^{38} shortfall undeniable. Demonstrating a known problem concretely is itself worth publishing.
4. **Distinguish derivation from fit, in writing.** A claim-status table that forces every number to be labelled *derived*, *coincidence*, or *falsified* is the cheapest defence against a coincidence (the mass ladder, the $\mathcal{O}(1)$ prefactors) being quietly promoted to a derivation.
5. **Pre-registration matters most without referees.** A hard negative-results rule and pre-registered decision rules are what let an independent programme falsify itself rather than accumulate suggestive positives. This is the part of SSV most worth copying.

9 Conclusion

SSV set out to derive nearly everything from one equation and two inputs. It ends having derived the *forms* of analogue gravity (which were already known), having needed three inputs rather than two once G was conceded, and having produced two clean negatives and one reusable numerical result. Measured against its ambition that is a failure; measured against the negative-results rule it adopted, the failure is the contribution. The lessons — forms are cheap and constants are dear, negatives are the durable output, concreteness exposes the magnitude gap, and self-imposed pre-registration is what makes honest closure possible — are not specific to a superfluid vacuum. They are addressed to anyone who tries to grow gravity from a medium.

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